

Evaluation of a Trust-Enabled Framework for Genomic Data Sharing within an Insurance Context: A Factorial Vignette Study

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Abstract

Genomics is revolutionizing the world we live, with the ability to sequence whole genomes rapidly and cheaply resulting in significant reductions in healthcare costs. Personalised medicine has the potential to contribute to the genomic revolution, but the current zero-sum game between protecting the data of participants and the expected gains from the research must be overcome. This research, using a factorial quantitative vignette study, evaluates the willingness of individuals to share their genomic data under a number of differing scenarios. Through the analysis of this survey, a trust-enabled framework comprised of high levels of Control, Transparency and Reciprocity is proposed to be implemented in the insurance industry, which if adopted would contribute significantly to genomic research. An examination of the influence of age and gender on an individual's willingness to share their genomic data is completed which offers insight into the decisions health insurers must consider before offering genomic testing to its policyholders. Furthermore, the research finds evidence to indicate that the results from the 2021 Edelman Trust Barometer, that there is a significant gap in trust between the informed public and mass population is not present in the insurance industry. To the best of the author's knowledge, a vignette experiment which investigates the preference of individuals within an insurance context has not been previously undertaken.

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1.0 Introduction

Genomics and the ability to sequence whole genomes rapidly and cheaply have brought about benefits to patients, society and the economy. This is evident through the recent significant reduction in healthcare costs across several clinical applications (Sabatini *et al.* 2016; Pennell *et al.* 2019). Genomics allows insight into more complex diseases which stem from a combination of genetic and environmental factors which facilitated the introduction of personalized and precision medicine (Seoighe *et al.* 2020). These recent advancements in technology have left us in a period where genetic information for research, clinical care and personal curiosity is becoming increasingly more prevalent. The sharing of this data is critical to enable progress for medical research and preventative care (Erlich and Narayanan 2014**).** Van Zyl (2017) analyzed the relationship which exists between individuals who avail of genomic tests and the insurance industry. This report examined the benefits insurance consumers would require to partake in Personal Genetic Testing (PTC) (Van Zyl 2017). The study highlighted the need for health and life insurers to seek both a prudent and ethical approach to genomic revolution to overcome the current zero-sum game that exists between the appetite to include an individual's genetic data in its underwriting procedures and the need for regulators to avoid genetic discrimination (Van Zyl 2017). This research proposes a trust-enabled framework to be implemented in the insurance industry which can contribute towards the genomic revolution, a framework first introduced by Erlich *et al.* (2014). This framework introduces three principles that facilitate trust in genetic research; Control, Transparency and Reciprocity (Erlich *et al.* 2014). This research finds that high levels of Transparency and Reciprocity, as well as dynamic consent framework for Control, are the conditions desired for policyholders to be most willing to share their genomic data. Furthermore, this trust-centric framework aims to act as a sustainable solution that brings together genetic privacy with data sharing and facilitates genetic research. Moreover, this study was unable to find evidence to support the findings from the 2021 Edelman Trust Barometer that there is a significant gap in trust between the informed public and the mass population present in the insurance industry. The mass population is 16 points less trusting than the informed public globally. As the majority of the respondents in the survey were residents of Ireland, it is important to note that the levels of trust inequality in Ireland is very similar to that of the global levels of trust, with Ireland experiencing a gap of 15 points (Edelman 2021). Presently, Ireland has failed to keep pace in both research and clinical application of modern genomics when compared to the US, China, Japan and other European countries (Stark *et al.* 2019). Seoighe *et al.* (2020) outlined

several measures which could be introduced to allow Ireland keep pace with other developed nations and contribute to the genomic revolution. The importance of recognizing the patient as the centre of the genomic revolution through the implementation of a nationwide public engagement and involvement programme for genomics was a key recommendation put forward in this report. This engagement would build a "social contract" of trust among the public and institutions (Seoighe *et al.* 2020).

The trust-centric framework outlined in this research offers health-insurers guidance on future decisions relating to the use of genomic testing on their customers. Pugnetti and Seitz (2021) evaluates how customer data can positively impact the insurance industry, with insurers' ability to improve decision making resulting in greater prevention and risk management. This study suggests that this can lead to improved pricing and significant reduction in losses for insurers (Pugnetti and Seitz 2021). Healthcare providers have previously demonstrated their ability to adapt the dynamics of their industry and respond to changes in the industry through effective decision making. This was highlighted in the period following the implementation of the Affordable Care Act in 2010, where healthcare stakeholders, including health insurers had to adapt to market trends such as patients becoming more informed and the rise in popularity of alternative provider payment models (Santilli and Vogenberg 2015). This research highlights an initiative through the implementation of a trust-enabled framework, how health insurers could adapt their strategies to consider the policyholder in future decisions, to keep pace with the genomic revolution currently evident in the insurance industry.

A recent report released by The House of Commons and Science and Technology committee has called the UK government to update existing regulations for direct to consumer (DTC) genetic and genomic tests (Brice 2021). One of the key recommendations put forward in this report was to consider the revision of regulation regarding the advice and support offered when DTC genomic testing is undertaken, including a requirement for the provision of genetic counselling (Brice 2021). This study offers guidance on how people's willingness to share their genomic data is influenced by the provision of genetic counselling and offers insight to policymakers into the importance individuals place on receiving genetic counselling.

1.1 Research Aims

This research aims to examine Erlich *et al.* (2014)'s trust-enabled framework and adapt it for an insurance perspective. The research will examine the belief that there is a positive relationship between an individual's willingness to share their genomic data and the levels of

Transparency, Control and Reciprocity they receive. Secondly, the research aims to determine if there is evidence to support the latest findings that there are record levels of trust inequality between the informed public and the mass population, which was recently reported in the 2021 Edelman Trust Barometer. Edelman is an organization which globally measures the level of trust towards institutions including Governments, Media Outlets, NGO's and Businesses (Edelman 2021). Thirdly, the research aims to establish if an individual's demographic characteristics influences their willingness to share their genomic data with insurance firms. Haeusermann *et al.* (2018) found that the right to privacy, even if formally guaranteed, is not enjoyed by all individuals in equal measures. This research aims to examine if this unequal attitude among various demographic groups towards privacy is evident within genomic data sharing in the insurance context.

This research contributes to the literature concerning an individual's willingness to share their genomic data with insurance firms by providing consumers with the opportunity to undertake genomic tests where genetic discrimination is not involved. This is undertaken through a factorial quantitative vignette experiment. The research aims to contribute to genomic research by developing an appropriate framework for genomic data sharing. To the best of the author's knowledge, a quantitative factorial vignette experiment has never been used to examine individual's date-sharing preference within an insurance context. Therefore, the originality of this paper is evident through the adoption of different vignette factors and the implementation of a genomic data-sharing framework in an insurance context. This study provides a recommendation for an appropriate trust-enabled framework to be developed, which could enable insurance companies to truly support modern genomics research.

1.2 Structure

The research paper is constructed as follows. Section 2 completes a review of the literature surrounding the current shift towards personalized medicine. An overview of individuals attitudes towards genomic data sharing will also be examined in this section. Additionally, the growth of genomics within the health insurance industry is discussed. This section concludes with an introduction to the vignette methodology used in this paper and discusses previous research which utilized this methodology. Further details on the development of this methodology and on its implementation are described in Section 3. Section 4 analyzes the findings in three steps. Firstly, the descriptive statistics of each dependent variable are provided. A regression analysis is then completed on the dependent variables with the vignette dimensions be applied as predictors. This section also examines how respondent characteristics

impact the willingness of individuals to share their genomic data. Section 5 concludes with a discussion of these results and provides a summary of the findings while also noting the research limitations. The conclusion offers insight into the impact the findings of this study have on the insurance industry, while also offering providing avenues for future research.

2.0 Literature Review

The 100,000 Genome Project, the Genomic Medicine 2025 and the Precision Medicine Initiative in the UK, France and the US, respectively, contributed significantly towards the shift in favour of personalized medicine. This section firstly examines these initiatives, and then proceeds to examine how attitudes towards genomic data sharing has been developing in recent years. The extent to which the literature has found an individual's characteristics can influence their outlook on genomic data sharing is detailed. Additionally, the risk of rising levels of asymmetric information within the insurance industry is highlighted, with current efforts made to combat this risk in Switzerland, the UK and Australia discussed. The section concludes with an examination of studies that have implemented a factorial vignette experiment, with insight offered on the benefits and limitations of this methodology.

2.1 Shift towards Personalized Medicine

Several studies have labelled personalized medicine through the use of genome and exome sequencing as the revolution that could transform how healthcare will be delivered in the future (Hong *et al.* 2013; Delaney *et al.* 2016; Morganti *et al.* 2019). At present, its usage is limited to a small percentage of patients, primarily for diagnosing suspected Mendelian conditions and cancer treatments (Delaney *et al.* 2016). The availability of genomic data from large numbers of individuals enables greater investigations to be undertaken to determine relationships and causes of rare and common diseases in different populations while also offering more information for understanding clinical care outcomes for an individual. In the past few years, several large-scale genome sequencing initiatives have been implemented in efforts to achieve this potential. The UK launched the 100,000 Genomes Project, which analyzed the genome sequences of individuals diagnosed with rare diseases and cancer. France have pledged to invest €670 million in a plan known as Genomic Medicine 2025, which is a genomics and personalized medicine programme (Lévy 2016; Horgan *et al.* 2017). The US aims to gather over one million volunteers to enable personalized medicine to be extended to all diseases in the Precision Medicine Initiative. Finally, China has plans to develop its own precision medicine initiative with USD 10 billion been pledged by the government (Horgan *et al.* 2017).

The literature notes that the integration of personalized medicine into everyday practice requires technological systems which can record genomic information in a clinically meaningful manner, and in a method that evokes confidence and trust with the public. These systems have yet to be appropriately developed (Delaney *et al.* 2016). Moreover, Pot *et al.* (2020) noted that in order for the successful integration of personalized medicine to be achieved, that there must be benefits for all stakeholders, including patients, healthcare providers and the healthcare system.

2.2 Attitudes towards genomic data sharing

To achieve the effective advancement of genomic research, the broader dissemination of genetic tests alone will not be sufficient. Instead, human participation in sharing genomic data in both public and private large-scale international genomic datasets and biobanks is critical to the success of the future of genomic research (Brookes and Robinson 2015; Kinkorová 2016). The views of participants towards the use of this data sharing is crucial for future development of research (Goodman *et al.* 2017). The literature has examined possible factors which influence participants' willingness to share their data. Trinidad *et al.* (2010) concluded that participants believe de-identified data should be accessible to the research community as it is considered to promote a societal goal. However, participants were less willing to share data with for-profit organizations (Trinidad *et al.* 2010). This supports research that has consistently highlighted that participants are less willing to share data with for-profit companies than with university researchers (Pentz *et al.* 2006; Helft *et al.* 2007; Kaufman *et al.* 2009). Though, more recent literature has opposed these findings. For example, Briscoe *et al.* (2020) found that an individual's perception of the risks and benefits of providing genomic data has little correlation with the sector making the data request. The research noted that organizations often associated with stronger societal missions, such as universities or hospitals, had only a slightly greater response to 'donor' rates than that of commercial businesses. Additionally, the governance policy of the organizations had no influence on participants willingness to share their data (Briscoe *et al.* 2020). Mello *et al.* (2018) found that participants are now likely to share genomic data with for-profit companies. Their study highlighted that 82% of respondents were somewhat willing to share data with for-profit companies, while respondents most significant concern with sharing their data was the fear their data would be used for marketing purposes (34%) or stolen (30%) (Mello *et al.* 2018). This research presents a number of hypothetical questions to individuals about their willingness to share their data. Previous research

undertaken in Indiana adopted this method where 88.4% of 274 cancer patients were willing to share their tissue samples for research under any condition (Nanibaa'A *et al.* 2016).

The literature examines how an individual's perception towards regulation in the field has hindered the growth in genomic testing. Legislation has been developed, such as The Genetic Information for Non-Discrimination Act (GINA) to combat these concerns. GINA prohibits the use of genetic information in health insurance when determining the premium and eligibility of an individual (Peterson *et al.* 2018). Despite the introduction of regulation, the potential for discrimination remains a significant barrier to genetic testing and sharing of data (Sharaf *et al.* 2013). Additionally, concerns regarding individual's privacy have influenced individual's willingness to share their genomic information. These concerns are justifiable, as there have been several instances where the failure of de-identification techniques highlighted that appropriate data protection could not be guaranteed in certain circumstances (Gymrek *et al.* 2013; Shringarpure and Bustamante 2015).

2.3 Genomics within the insurance industry

Previous studies have examined the implications to life and health insurers when individuals obtain results from genetic tests. Issues surrounding asymmetric information have been examined as a key concern for healthcare providers, with individuals being able to gather additional information on their susceptibilities to certain illnesses and diseases. Insurers argue that they need to obtain any knowledge that may enhance the predictability of an individual being exposed to a disease (Bombard *et al.* 2010). Without appropriate access to this genetic data, insurers expect consumers to purchase greater levels of insurance, at a price that may not reflect their true risk, which would negatively impact the sustainability of the industry (Bombard *et al.* 2010). Prior literature has examined several unique methods which attempt to combat this issue of asymmetric information. Switzerland, the UK and Australia have imposed a monetary cap for policies where genetic data may be used. This allows insurers to hedge against risk for their most expensive policies, allowing firms to distribute risk on policyholders if significant net-worth is opposed to the vulnerable (Bélisle-Pipon *et al.* 2019). There has been several studies which have established that individuals who have been exposed to their genetic information and have been found to have a greater exposure of developing a particular disease, are more likely to alter their insurance policies (Zick *et al.* 2005; Chao *et al.* 2008). Therefore, as the availability of genetic testing increases, it is likely it will have a negative impact on insurers, with the impacts growing proportionally as progression in the field of genomic research continues (Howard and FCIA 2014).

2.4 Factorial Vignette Methodology

The use of a factorial quantitative vignette study is the primary research method used in this research. The literature has extensively described the appropriate modes to construct and implement quantitative vignette studies (Hughes and Huby 2004; Ganong and Coleman 2006; Atzmüller and Steiner 2010). Vignette studies use several short scenarios (vignettes) presented to respondents within surveys to extract their judgements relating to the situations (Atzmüller and Steiner 2010). Previous studies note this research method is constructed using realistic scenarios to access dependent variables of participants (Aguinis and Bradley 2014). This enhances experimental realism while offering academics an opportunity to manipulate and Control independent variables (Ibid). Erfanian *et al.* (2020) document the advantages of vignettes in research, noting that they are very flexible in nature and are effective in targeting specific complex issues. Moreover, the use of standardized social stimulus across all volunteers results in increased internal validity, reliability and easily replicable (Alexander and Becker 1978; Atzmüller and Steiner 2010). The detachment of participants' experience from vignettes characters, and the debate between surrounding belief and action of participants are found to be drawbacks to this form of research (Erfanian *et al.* 2020).

This research implements a factorial vignette study, which combines survey approaches with an experimental research design (Aviram 2012). Factorial vignette surveys have previously used several specific research areas such as public policy (Moffat *et al.* 2016), medical research (Soon *et al.* 2019), human behaviour research (Young *et al.* 2017), social experiment research (Carlson 1999), judicial research (Reisig *et al.* 2018; Flippin *et al.* 2019) and business ethics research (Martin 2012). The randomized component included in the survey results in each respondent presented with 5 out of the 12 vignettes. This is similar to the method adopted by Soon *et al.* (2019), where participants are randomly allocated one of the four vignettes, while Nivette and Akoensi (2019) present respondents with 2 of 8 vignettes. Factorial vignette design aids to define the independent contribution of several variables to a respondent's judgement without any respondent required to answer all possible vignette scenarios.

3.0 Methodology

This section of the paper outlines the distribution strategy, the construction of the quantitative vignette survey and the selection of vignettes to be presented to respondents.

3.1 Distribution of Survey

A research aim of this study is to determine if there is evidence to support the latest findings from Edelman (2021) that there are record levels of trust inequality between the informed public and the mass population within the insurance industry. An individual must be aged between 25-64, college-educated, in the top 25% of household income per age group and must also actively engage in public policy and businesses news to be part of the informed public (Edelman 2021). Therefore, the strategy of this research is to distribute the survey to individuals over the age of 18, and through the initial questions in the demographics section of the questionnaire, each respondent will be classified as either an informed individual or part of the mass population. The survey was accessible to individuals from all countries as this will allow for a comparison to be made among the levels of trust within different nations. The survey was initially distributed to postgraduate students at the University of Limerick to act as a pilot for the research. Following this testing, the survey was distributed across social media platforms where all connections over 18 years of age were invited to participate.

The literature does not indicate if there is appropriate sample size for a factorial vignette survey. An appropriate sample should consider the number of factors and factor levels and the number of vignettes (Weber 1992). Weber (1992)'s review noted that many researchers considered 100 subjects on a common sample size to be sufficient. However, Tilburt *et al.* (2010)'s factorial vignette survey believed 1200 participants to be an appropriate sample size. Therefore, this research believes an appropriate number of respondents lies between these two studies and aims to attract 250-300 respondents, with each vignette to be completed approximately 100 times.

3.2 Factorial Vignette Experiment

This study adopts a vignette-based factorial survey to establish a trust-enabled framework for genomic data sharing within an insurance context. This form of survey is particularly beneficial when the topic being investigated is of a sensitive manner (Aviram 2012; Aguinis and Bradley 2014). When factorial surveys are compared to traditional item-based statements, they are found to act as a more realistic setting and as its purpose is less obvious to respondents, it can

lead to greater levels of honest answers and also lowers the risks of social desirability[1] (Dickel and Graeff 2018; Oll *et al.* 2018).

The initial step in developing a vignette experiment requires the appropriate selection of factors and factor levels that are then factorially combined to produce the entire vignette population (Su and Steiner 2020). Failure to include important variables in complex decision-making processes can result in the validity of the study been undermined (Aguinis and Bradley 2014). For this research, each vignette represents a different proposal which is presented by the insurer to the policyholder, with each vignette providing a different incentive to the individual to share their genomic data. Each of the scenarios presented to the policyholders is based upon two assumptions. Firstly, that insurers will not use an individual's genomic data for underwriting purposes, although the future insurability of a participant could be affected depending on laws within their own country. Consequently, participants will not be subject to genetic discrimination for partaking in testing. Secondly, the de-identified genomic information provided by individuals may be used, shared, or commercialized for research purposes, depending on the conditions outlined in each scenario. These assumptions are built from a programme offered by Discovery, a South African insurance firm that offered their customers the opportunity to partake in genetic analysis at a discounted price, in return participants were presented with a comprehensive report of their individual disease risks as well as personalized strategies to enable policyholders to improve their health (Cookson 2015).

The twelve vignettes presented to respondents vary with respect to the three principles outlined by Erlich *et al.* (2014), which are aimed to build and maintain trust between participants and researchers. Each vignette is composed of differing levels of Transparency, Control and Reciprocity. Consistent with this studies definitions, three factor levels are adopted in relation to Transparency (zero Transparency, intended use and actual use), and two factor levels are considered with relation to Control (one-time decision and dynamic consent framework). Reciprocity relates to the extent in which the research community "pays back" a volunteer for contributing towards science. The author notes how a sense of community among participants can enable the gap between societal and individual rewards to be bridged (Erlich *et al.* 2014). Therefore, this research has adopted this belief to an insurance context by using three factors for Reciprocity, with the lowest level consisting of an individual receiving a report of their personal health. If an insurance firm is presenting their policyholders the opportunity to

[1] "Social Desirability" occurs when respondents tend to admit to traits and behaviours which can be seen as socially desirable and deny traits and behaviours which are socially undesirable (Krumpal 2013).

undergo genomic tests with the intent of aiding them to live longer and healthier lives, providing them with the results of their personal health must be considered a minimum requirement for volunteers. The medium level of Reciprocity relates to the addition of facilitated access to genetic counselling in order to promote disease prevention. The highest Reciprocity level to be adopted in the research consists of the addition of an economic reward to participants. Previous research has highlighted that financial rewards can incentivize individuals to share their data. Roberts *et al.* (2017) note how the ability for an individual to profit from their own genetic data facilitates the opportunity to gain from the sharing of this information while also feeling they have made an important and unique contribution to scientific research. The reimbursement of the cost of being tested acts as the highest level of Reciprocity in this research, where this level includes receiving a report of personal health, gaining access to genetic counselling and the cost of the testing being paid by the insurance company. This study considers this level of Reciprocity to be a realistic option for insurance firms, particularly as the cost genome sequencing decreases into the future, and the opportunity for whole genome sequencing becomes more readily available for research purposes, as noted by Delaney *et al.* (2016). Moreover, it allows for an examination of how the cost of testing could act as a potential discriminating factor for an individual who agrees to undergo genomic testing.

The full population of vignettes is made up of 18 different scenarios (3 x 2 x 3). However, the research considers six combinations of the vignettes to be unrealistic and are subsequently removed. The author considers the combination of a high level of Control with both a low and medium level of Transparency can be considered unrealistic, as the opportunity to provide dynamic consent is reliant on notification of the actual potential use of the data. The decision to remove unreasonable vignettes is supported by Wason *et al.* (2002), where the authors note that unreasonable vignettes must be removed and that all vignettes must be believable, even if they extend beyond the experience of the respondents.

Once the unrealistic vignettes are removed, there are 12 vignettes remaining which comprise the final set of vignettes. The survey was generated on Qualtrics software which is an online survey tool. Each participant was randomly assigned 5 of the 12 vignettes, the platform allowed for each vignette to be evenly presented to respondents to ensure all scenarios have been equally reviewed. Sauer *et al.* (2011) completed an examination on the effect the number of dimensions and vignettes has on the consistency of findings. The paper noted that when respondents are part of the general population, lower educational respondents show greater

levels of inconsistency in their evaluations when the number of vignettes was high. The paper recommended that when factorial surveys are being applied to general population samples, a limited number of vignettes and dimensions per respondent should be presented. Therefore, this research believes 5 out of the 12 vignettes is appropriate per respondent. In order for the statistical analysis of a vignette study to be valid, each vignette must be measured an equal number of times (Weber 1992). Additionally, this research adopts the approach described by Alexander and Becker (1978), which randomly allocates different versions of the same basic vignette to different participants, which attempts to remove differences in ethical judgements that could influence the rating process independently from the selected factors. This randomization within the vignettes and in the allocation of vignettes offers this factorial survey the robustness of an experimental method.

3.3 Traditional Survey

The initial section of the questionnaire is comprised of questions that determine the demographic characteristics of the participant. Shabani *et al.* (2014) completed a review of previous surveys which examined individual's perception of genomic data sharing. The paper noted that characteristics including race, gender, age, marital status, and education level could impact an individual's decision on whether to share their genomic data. Therefore, in the initial section of the questionnaire, participants are asked to answer questions relating to their personal demographics. Moreover, questions relating to average household income and levels of engagement with both business and public policy news are included to determine if a respondent can be classified as a member of the "informed public" according to Edelman (2021)'s standard. Each respondent can then be classed as either a member of the "informed public" or "mass population". A respondent can only be a member of one group.

3.4 Data Analysis

Respondents' willingness to share their genomic data in each scenario was established through an 11-point Likert scale ranging from 'strongly disagree' [0] to 'strongly agree' [10]. Firstly, to provide a generalized overview of the findings, a table is generated that highlights the average response to each vignette of individuals' willingness to share their genomic data. Each vignette's data is presented in terms of the entire population of respondents, the informed public and the mass population. Moreover, the relationship between specific demographic characteristics and individual's willingness to share their data is presented. This research examines the correlation coefficient to assess the relationship between these characteristics. For these analyses to be completed, each of the independent variables (Control, Transparency

& Reciprocity) were expressed as a binary (dummy) variable. The resulting coefficients are then examined to evaluate the extent in which each of the levels of the independent variables influence the willingness of an individual to share their genomic data.

4.0 Results

321 individuals clicked on the survey, with 228 respondents completing the survey, which results in a completion rate of 71%. The analysis of these respondents is completed in three steps. Initially, the descriptive statistics of each of the dependent variables are described. Secondly, a regression analysis is completed on the dependent variables applying the vignette dimensions as predictors**.** Thirdly, the respondent characteristics (age, gender, informed public) are included and described within the regression analysis.

4.1 Descriptive Statistics

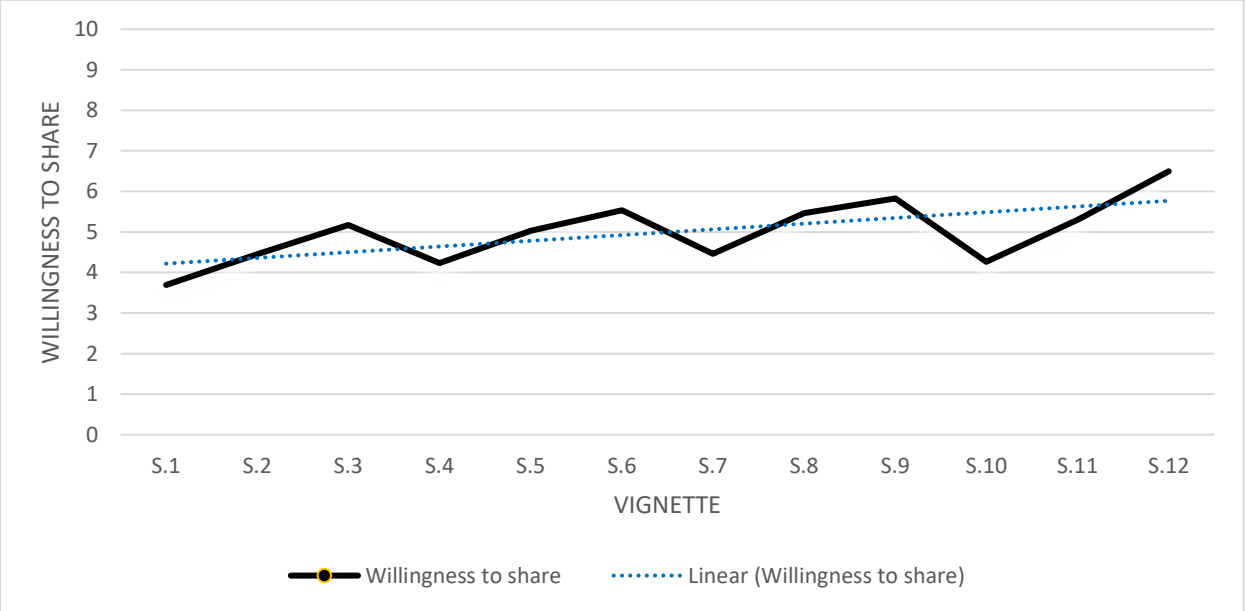
The initial analysis of the study completes an examination of the means and frequency of distribution of individual's willingness to share their genomic data depending on the vignette conditions. The unit of analysis is the response made to each vignette, rather than the subject who completed the response, with each response completed by respondents treated as independent. The research acknowledges that a certain level of intra-respondent correlation may occur, as discussed by Nock and Rossi (1982). *Figure 1* illustrates the mean response to each vignette. It is evident from the trendline that there is a slight positive trend from left to right, which indicates that as higher levels of Control, Transparency and Reciprocity are added to vignettes, respondents become more willing to share their data. S.1, (one-Time decision, low Transparency, low Reciprocity) scored the lowest in response with a mean response of 3.69. This vignette factor levels contained the lowest levels of Reciprocity and Transparency and a one-time decision in terms of Control of an individual's data. Contrastingly, vignette S.12 has the highest willingness of participants to share their genomic data with a score 6.50. This vignette factor levels are comprised of the highest level of Transparency and Reciprocity, while also contains a dynamic consent framework.

The frequency distribution indicates that each of the vignettes lies within the middle section of the scale (4-6), which suggests that despite each vignette been comprised of varying factors levels of Control, Transparency and Reciprocity, individuals remain uncertain surrounding their view towards sharing their data.

Table 1: Descriptive Statistics

Vignette	Nr of Respondents	Control	Transparency	Reciprocity	Mean	Std Dev
S.1	97	O	Low	Low	3.69	2.62
S.2	99	O	Low	Medium	4.45	2.75
S.3	102	O	Low	High	5.17	2.79
S.4	101	O	Medium	Low	4.23	2
S.5	101	O	Medium	Medium	5.03	2.7
S.6	103	O	Medium	High	5.53	2.91
S.7	102	D	Medium	Low	4.46	2.55
S.8	99	D	Medium	Medium	5.46	2.69
S.9	102	D	Medium	High	5.82	2.79
S.10	95	D	High	Low	4.26	2.54
S.11	101	D	High	Medium	5.3	2.76
S.12	101	D	High	High	6.5	2.59
*O = One-Time Decision						
*D = Dynamic Decision Making						

Figure 1: Visualisation of Average Willingness to Share Genomic Data



4.2. Regression Analysis

The initial step of analyzing factorial surveys is the study of a basic theoretical model, which examines the effect in which the factors in the vignettes have on the decision (Taylor 2006). Multiple linear regression is the statistical method adopted in this research to describe and measure the relationship between each independent variable (varying levels of Control, Transparency, and Reciprocity) and the decision made (individual's willingness to share their genomic data). This method makes it possible to infer a causal relationship, which means it can be determined if each independent variable influences the change in the decision, or whether the variable is only associated with the decision by "accident" (Taylor 2006). This can be done as the factors in the vignette are deemed independent. *Table 2* outlines the findings from this analysis.

The initial multiple regression analysis has an R-squared value of 0.975, indicating that independent variables can explain 97.5% of the variance in the dependent variable. Additionally, the model can be considered statistically significant as its F significance has a p-value of 0.000713, which is significantly lower than the 99% confidence value of 0.01.

Firstly, the initial vignette level of a one-time informed decision is set as the baseline level in terms of Control. When the vignette contained the dynamic consent level of Control instead of the one-time decision level, an individual's willingness to share their genomic data would increase by 0.319 units (i.e., if an individual's average response to share their genomic data was 4 where a one-time decision was a condition present, it can be expected that the average response would increase to 4.319 if all other conditions remained the same). However, this finding can be considered to be statistically insignificant due to the p-value score of 0.151 which fails to lie within the 95% confidence levels. The second vignette factor to be addressed relates to varying levels of Transparency. The lowest level of Transparency where no further level of information is provided to the respondent once permission is granted is set as the baseline level. The output from the regression analysis established that an individual is 0.493 and 0.595 more willing to share their genomic data for the medium and high levels of Transparency, respectively. Similar to the Control factor level, the p-value for the highest level of Transparency is greater than 0.05, at a value of 0.073, indicating that the study cannot say with a confidence level of 95% that the correlation is statistically significant. The p-value of 0.044 for the medium level of Transparency results in this research accepting with a 95% confidence interval that as the level of Transparency rises from low to medium, that an individual's willingness to share their genomic data increases by a value of 0.493+-(S.D). This

implies that if a scenario was presented to individuals to share their genomic data resulted in a mean response of 6, this study accepts with a 95% confidence level that if the Transparency condition was changed from the lowest level to medium level of Transparency, that the response would increase to 6.493+-(0.194). Thirdly, when Reciprocity is examined, the lowest level of Reciprocity where an individual receives only their genomic results for participating in genomic testing is set as the base level for the analysis. The output from the regression analysis resulted in coefficients of 0.901 and 1.594 with the base level, and the medium and high levels of Reciprocity respectively. The p-values for these variables are considered statistically significant at a 99% confidence interval (0.0017, 0.000078).

Table 2: Regression of Willingness to share genomic data on vignette factors and levels

	Coefficients	**Standard Deviation**	**P-Value**
Control*			
Dynamic Consent	0.319	0.194	0.151
Transparency **			
Medium Transparency	0.493	0.194	0.04399
High Transparency	0.595	0.274	0.07315
Reciprocity* **			
Medium Reciprocity	0.901	0.168	0.00173
High Reciprocity	1.594	0.168	7.815E-05

**One-Time Decision set as Base Level for Control*

*** Low Transparency set as Base Level for Transparency*

**** Low Reciprocity set as Base Level for Reciprocity*

4.3 Demographics

Determining if there is an unequal attitude among various demographic groups towards genomic data sharing within the insurance context is one of this research aims. The following section examines findings from the survey by analysing both respondents' gender and age.

4.3.1 Gender

Table 3 presents a comparison between male and female respondents. In 11 out of the 12 vignettes presented to respondents, it is found that males are more willing to share their genomic data with insurance firms than females are. This supports the literature where Goddard *et al.* (2009) noted that males were 11% more likely to contribute to genomic research through biobanks than females. S.4 is the vignette with the most significant difference in willingness to share data between males and females. When this response is compared to S.7, which has the same levels of Transparency and Reciprocity but differs in decision of Control from a one-time decision to a dynamic consent framework, female's willingness to share increases by 8.7% (up from 3.91 to 4.25). Males' willingness to share their data rises slightly by 4%, from 4.48 to 4.62. This indicates that females place a greater importance on the Control of sharing their data than their male counterparts. Moreover, S.12 has a significant gap between the willingness of males' vs females to share their genomic data (6.26 v 6.77). S.9 and S.12 have similar conditions, with both containing a dynamic consent framework of Control and the highest level of Reciprocity, however S.9 has medium level of Transparency, which is one level lower than that in S.12. Females realize a rise in willingness to share their genomic data by 16% (5.79 to 6.77), whiles males see a rise of 7% (6.26 to 6.77). This suggests that females place more importance on the Transparency of what their data will be used for than their male counterparts.

Table 3: Comparison of Male V Female Responses to each Vignette

VIGNETTE	MALE COUNT	MALE RESPONSE	FEMALE COUNT	FEMALE RESPONSE
S.1	54	4	42	3.24
S.2	58	4.69	41	4.12
S.3	60	5.32	41	4.83
S.4	56	4.48	45	3.91
S.5	64	5.11	37	4.89
S.6	60	5.77	43	5.21
S.7	58	4.62	44	4.25
S.8	55	5.62	44	5.27
S.9	60	5.85	42	5.79
S.10	60	4.27	34	4.09
S.11	55	5.36	45	5.11
S.12	61	6.26	39	6.77

4.3.2 Age

The research examined the relationship between a respondent's age, and their willingness to share their genomic data. It is found that in 9 of the 12 vignettes that there was a negative correlation between an individual's age and their willingness to share their data. This indicates the older the respondent, the less willing they are to share their genomic data, however several of the correlations are weak and fail to offer any significant insight into the relationships between an individual's age and their willingness to share their data. Goodman *et al.* (2017) previously found that the older the age of respondent was significantly associated with a decreased willingness to share genomic data for research purposes. S.1, S.2 & S.3 show evidence to support these findings. Each of these vignettes are comprised of low levels of Control and Transparency, but levels of Reciprocity vary. When these correlations are considered with the vignettes where levels of Transparency are altered to include the intended use and aims of projects utilizing the data (S.4, S.5, S.6), while all else remaining equal this negative correlation between age and willingness to share data diminishes. This indicates that older adults are less willing to share their genomic data than younger adults when the use of the data is not very transparent to the participant. Table 3 documents the correlation of each vignette. S.12 has the highest positive correlations between age and willingness to share with a correlation coefficient of 0.0912. When this is considered with S.9 (-0.932), where the only varying factor level is the change in Transparency from a medium level to the highest level of Transparency, this further supports the belief that older adults place a significant importance on levels of Transparency when considering sharing their genomic data.

Table 4: Correlation of Age/Willingness to share data

VIGNETTE	CORRELATION (AGE/RESPONSE)
S.1	-0.3007
S.2	-0.3619
S.3	-0.1551
S.4	0.0194
S.5	-0.0041
S.6	-0.02316
S.7	-0.0409
S.8	0.0147
S.9	-0.0932
S.10	-0.0642
S.11	-0.119
S.12	0.0912

4.4 Mass Population v Informed Public

The 2021 Edelman Trust barometer has reported that there are record levels of trust inequality between the informed public and the mass population. In Ireland, there is a 15-point gap between the two groups. This research does not have evidence to support these findings. However, further respondents are required in order to determine the statistical relevance of these findings. *Table 5* documents the current findings.

Table 5: Mass Population V Informed Public

Vignette	Mass Population Respondents	Average Response	Informed Respondents	Average Response	Difference in Response
S.1	77	4.09	20	2.15	**1.94**
S.2	71	4.92	28	3.29	**1.63**
S.3	77	5.44	25	4.32	**1.12**
S.4	74	4.24	27	4.19	**0.06**
S.5	81	4.84	20	5.80	**-0.96**
S.6	79	5.52	24	5.58	**-0.06**
S.7	85	4.34	17	5.06	**-0.72**
S.8	80	5.45	19	5.53	**-0.08**
S.9	82	5.57	20	6.85	**-1.28**
S.10	70	4.24	25	4.32	**-0.08**
S.11	78	5.23	23	5.52	**-0.29**
S.12	78	6.47	23	6.57	**-0.09**

As discussed in the previous section, this research was unable to support the latest findings released by Edelman in 2021 that a significant gap in trust exists between the mass population and the informed public. Further statistical analysis is undertaken on the data to establish the relationship between the mass population, the informed public and an individual's willingness to share their data. *Table 6* highlights the results from linear regression analysis completed on

each of the vignettes. While 10 of the 12 regressions completed were found to be statistically irrelevant, tests completed on S.1 and S.2 offer appropriate statistical insight into the data. The mass population has a positive coefficient of 1.94 and 1.629 per unit of willingness to share their genomic data over the informed public, with both vignettes having of a confidence interval of greater than 99%. This indicates for these two vignettes that members of the mass population are more willing to share their genomic data than members of the informed public. Both S1 and S2 have low levels of Control and Transparency but differ in relation to Reciprocity. S2 provides participants free facilitated access to genomic counselling in order to analyze the results and promote disease prevention, a service not provided in S1. This suggests that the mass population under these conditions are more willing to share their data, despite having low Control and Transparency over the data. This would oppose the findings in the Edelman Trust Barometer 2021 that members of the informed public have greater levels of trust in institutions than the mass population.

Table 6: Regression of Mass Population Effects on Individual's Willingness to Share Their Data

	Coefficient	Standard Error	P-Value
	Mass Population		
S.1	1.94	0.631	0.002749
S.2	1.630	0.594	0.007
S.3	1.12	0.6356	0.08
S.4	0.058	0.587	0.92
S.5	-0.960	0.672	0.156
S.6	-0.06435	0.682	0.930
S.7	-0.71765	0.676	0.291
S.8	-0.076	0.691	0.912
S.9	-1.27	0.689	0.0667
S.10	-0.077	0.595	0.897

S.11	-0.291	0.657	0.659
S.12	-0.091	0.619	0.884

*** Informed Public is set as Base Level for examination of relationships*

5.0 Discussion

This research has examined the relationship on how an individual's willingness to share their genomic data is impacted by varying levels of Control, Transparency and Reciprocity using a factorial vignette survey. Overall, participants ratings of willingness to share their data was quite low insofar that only one vignette exceeded a mean of 6 on the 10-point Likert scale. This finding can be viewed as a slightly surprising when considered with previous work in this field. Recently, Mello *et al.* (2018) discovered that 82% of their respondents were somewhat willing to share their data. This study failed to find strong evidence supporting this finding, with S.12 being the only vignette which a mean response which showed respondents were significantly more likely than not to share their genomic data. However, when the standard deviation of each vignette is considered, all vignettes lie within the range between 4-6 on the Likert Scale which fails to provide significant insight into the willingness of individuals to share their data as the vignette factor levels change. Further research should target a larger sample size in an attempt to gain more accurate mean responses to individual's attitude.

The regression analysis completed on the vignette factor levels offers significant insight to the insurance industry on what factors individuals find the most important when considering sharing their genomic data. These findings result in the recommendation of the development of a trust-enabled framework which incorporates Reciprocity at the centre of the framework. The most significant result from this analysis is the coefficient of both the medium and high Reciprocity, with 0.901 and 1.594 respectively. This indicates that when the provision of genomic counselling in order to analyze the results and promote disease prevention is provided to participants who share their data, that they would be .901 units more likely to participate than if this service was not provided. A further 1.594 units of willingness to share data is present when both genomic counselling and the cost of testing is covered. The p-values of 0.00173 and 7.815E-05 allows this research to accept this hypothesis at a 99% confidence level. This finding is in line with Turland and Slade (2020) findings which noted that the introduction of small financial and non-financial benefits significantly increase participants willingness to share data. Moreover, this analysis offers insight into the importance individuals place on varying

levels of Transparency. Particularly, this research can accept the hypothesis that individual's willingness to share their genomic data will increase by 0.493 +-0.194 when information about the institutions with access to the data and the intended use of the data is made known to participants, when compared to no details been provided once permission is granted. Furthermore, in terms of Transparency, correlation testing completed on the vignettes and age highlighted that older adults tend to place a greater importance on the levels of Transparency than younger adults. Previous research in this field supports these findings (Goddard *et al.* 2009; Shabani *et al.* 2014). Leon *et al.* (2013)'s research found that participants were far less willing to share data when information regarding the use their data was not transparent, however this study did not observe significant differences across demographic conditions for this finding. This research paper suggests that demographics does influence individuals' willingness to share data when levels of levels of transparency is varied. When the results of this regression analysis are considered with the findings from the descriptive statistics, this study proposes that health insurers should implement a framework where levels of Control, Transparency and Reciprocity are high, and if health insurers are to prioritise one factor above another, Reciprocity should be of most significant importance.

The research finds evidence to indicate that the latest Edelman (2021) findings are not present in the insurance industry. The regression analysis completed on each of the vignettes offered statistical insight into 2 of the 12 vignettes. When policyholders are offered to participate in genomic testing, and the conditions presented are of both low Control and Transparency, and low and medium levels of Reciprocity respectively, that individual's part of the mass population are more willing to share their genomic data than members of the informed public. This suggests that members of the mass population are more trusting in health insurers to obtain their data than members of the informed public, which opposes the findings outlined in Edelman's 2021 report.

5.1 Limitations

The author acknowledges several limitations of this study. A methodological limitation which is inherent in most surveys, is the problem of self-selection bias of respondents, where individuals who accept to complete the survey are often more willing to provide their data. Therefore, respondents may reflect individuals who are more positive and open towards the advancement of personalized health research through the sharing of their genomic data. Consequently, the true willingness of individuals to share their genomic data may be lower than this study suggests. Secondly, Aguinis and Bradley (2014) discuss that a common

challenge for vignette studies is that gaining access to an appropriate sample group can be difficult. This challenge is evident in this research, with most respondents falling within the young, uninformed public. However, as previously outlined, one of the objectives of this paper is to determine if there is evidence of trust inequality between the informed public and mass population. The paper was unable to attract sufficient members of the "informed public" with limited the extent in which this objective could be analyzed.

The ability to send respondents reminders to finish the questionnaire also acts as a significant limitation for the paper. Of the 321 individuals who began the survey, 93 failed to complete the questionnaire which represents 28.9% of the total respondents. The author has no means of establishing the reason for this failure to complete the survey. This may have resulted from a lack of interest, the vignettes being too complex, or a negative view towards the research. To overcome this limitation within future studies, it is recommended for a drop-out option to be included. Within this option, participants could describe their reason for exiting the questionnaire, which would provide clarity to the researcher for the reason for failing to complete the study. Furthermore, due to the manner of distribution, it was not possible to determine the response rate.

6.0 Conclusion and Opportunity for Future Research

Technological developments in recent years have led to the advancement of genomic research and rise in direct-to-consumer genetic testing. This has the potential to have a considerable impact on life and health insurers. Insurers are now considering offering consumers the chance to partake in genomic testing. The sharing of individuals genomic data is critical to the future progress of medical research and preventative care (Erlich and Narayanan 2014). This study believes that health insurers could significantly contribute to genomic research if the trust-enabled framework proposed in this research is adopted. The research provided evidence to support the first aim of this paper which was to examine the belief that there is a positive relationship between individual's willingness to share their genomic data and the levels of Transparency, Control and Reciprocity they receive. It was found that individuals are most willing to share their data when levels of Control, Transparency and Reciprocity are high. For health insurers considering providing genomic testing for their policyholders, it is important to note that individuals place greater emphasis on been "paid-back" by the research community, than the Transparency and Control they have off their data. Moreover, findings examined in this paper offer health insurers important discoveries in relation to how demographics influence

an individual's willingness to share their data. Significantly, older adults tend to place greater importance on the levels of Transparency than younger adults. Therefore, insurers must consider their portfolio of policyholders when deciding on the level of Transparency being provided to individuals when offering the opportunity to participate in genomic testing to ensure the greatest uptake in the initiative. The lack of evidence to support Edelman (2021)'s findings that there is a significant gap in trust between the mass population and the informed public will assist insurers if they are to implement the proposed trust-enabled framework. Insurers would not have to develop varying frameworks for each class of population, as there is no evidence to suggest the informed public would respond differently to the mass population when considering sharing their genomic data.

This study highlighted that levels of Transparency influence an individual's willingness to share their data. However, previous literature as seen in Nanibaa'A *et al.* (2016) review noted that 88.4% of participants were willing to share their data under any conditions, which indicated that Transparency was not the most important factor when undertaking genetic testing in this study. This study was completed on individuals who were cancer patients, which may have influenced their decision due to personal experience of the importance of genomic research. Future research could examine the relationship between individuals' health status and their willingness to share their genomic data. Furthermore, this study was completed under the assumption that an individual would not be subject to genetic discrimination for partaking in testing. Future research could complete a study in countries where stringent regulation in relation to use of such discrimination for underwriting purposes is not present.

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Appendices

Appendix A-Vignettes levels

S1: One-Time decision, Zero Transparency, Lowest level of Reciprocity

S.2: One-Time decision, Zero Transparency, Medium level of Reciprocity

s.3: One-Time decision, Zero Transparency, Highest level of Reciprocity

S.4: One-Time decision, Medium Transparency, Lowest level of Reciprocity

S.5: One-Time decision, Medium Transparency, Medium level of Reciprocity

S.6: One-Time decision, Medium Transparency, Highest level of Reciprocity

S.7: Dynamic consent framework, Medium Transparency, Lowest level of Reciprocity

S.8: Dynamic consent framework, Medium Transparency, Medium level of Reciprocity

S.9: Dynamic consent framework, Medium Transparency, Highest level of Reciprocity

S.10: Dynamic consent framework, High Transparency, Lowest level of Reciprocity

S.11: Dynamic consent framework, High Transparency, Medium level of Reciprocity

S.12: Dynamic consent framework, High Transparency, Highest level of Reciprocity

Appendix B-Information Sheet

Aims: The survey aims to evaluate a trust-enabled framework for genomic data sharing with an insurance company for research purposes.

Why have you been invited? If you are at least 18 years of age, you are welcome to partake in this research.

What will you have to do? You will be asked to evaluate a series of scenarios describing potential offers made by an insurance company in order for you to share your genomic data with them for research purposes. Moreover, you will be asked to answer some questions about yourself. If you wish to take part, simply click the "Yes" button below. By completing the online survey, you will be providing your informed consent to take part.

What will happen to your information? The information you provide is strictly confidential. For the purpose of the research, personal identifiable information (e. g. name, e-mail address) will not be required. The data will be aggregated, and the responses of individuals will not be identifiable. Moreover, the information provided is stored securely on the researchers' password protected computers. It can only be accessed by the researchers named below.

Do you have to take part? No. Participation is completely voluntary. If you change your mind once you started the survey, you can withdraw your participation at any stage during the survey. Simply close your browser window, and all your responses up to that point will be discarded.

Time commitment: Completing the survey will take no longer than 10 minutes. If you need to stop partway through responding to the survey, you can pause and return to the point later. This can be done by following the same link you used to access the survey and you will be returned to your most recent question completed.

What are the benefits for you? There are no known benefits for you personally. Your participation would help us to understand which factors are most relevant when sharing genomic data with an insurance company for research purposes.

What are the risks to you? There are no known risks of this study. However, if you do feel uncomfortable at any stage of this survey, please feel free to discontinue the questions by closing your web browser as outlined above. If you do this, any responses you have given up to that point will be discarded. If you have any queries about the study before taking part, please feel free to contact the researcher Páraic O'Loughlin (16179153@studentmail.ul.ie) or Dr. Barry Sheehan (Barry.Sheehan.ul.ie).

This research study has received ethical approval from the Kemmy Business School Research Ethics Committee. If you have any concerns about this study and wish to contact someone independent to the study, you may contact:

The Chair of KBSREC

Dr. Maria Lichrou,

University of Limerick,

Maria.Lichrou@ul.ie

Consent Do you agree to take part?

- ○ Yes
- ○ No

Appendix C : Part A – Questionnaire: Participant Characteristics & Demographics

Information

The section below asks questions regarding participant characteristics and demographics. The information you provide is strictly confidential. The data will be aggregated, and the responses of individual will not be identifiable.

Q1 Which of the following insurance products do you possess?

- ☐ Life Insurance
- ☐ Health Insurance
- ☐ Critical Illness Cover
- ☐ No product

Q2 What is your Country of Residence?

- ○ Ireland
- ○ England
- ○ Other (Please Specify) _____

Q3 What is your gender?

- ○ Male
- ○ Female
- ○ Non-binary
- ○ Prefer not to say

Q4 What is your age?

	0 10 20 30 40 50 60 70 80 90 100
Age?	

Q5 What is your marital status?

- Single, Never Married
- Married
- Separated or Divorced or Widowed

Q6 What is the highest degree or level of school you have completed?

- Less than Secondary School
- Secondary School
- Bachelor's degree or equivalent
- Master's degree or equivalent
- Doctoral degree or equivalent

Q7 Which of these best describes your average household income?

- ○ €0-€24999
- ○ €25000-€49999
- ○ €50000-€74999
- ○ €75000-€99999
- ○ €100000 and greater
- ○ Prefer not to say

Q8 Do you actively engage and keep up-to-date with both business and public policy news?

- ○ Yes
- ○ No

Appendix D: Part B- Vignette Experiment

Please read the following and assume it to be your current situation:

You currently own a health insurance policy and have recently received your renewal quote. With your quote you have been granted the opportunity to partake in genomic testing by your insurer.

Genomic testing is when an individual undergoes tests which establishes their susceptibility to various diseases based on their genetic make-up.

If you decide to participate in the testing, you will receive your personal genomic testing results, including information regarding your risk of contracting specific diseases and you will contribute to genomic research as your data will be used or shared with third parties (in a deidentified form) exclusively for research purposes. The highest standards of data anonymization and security will be implemented to protect your privacy. Re-identification through genomic data is highly unlikely but cannot be guaranteed. Moreover, according to regulations in force, sharing your data won't affect your access to insurance products, either through price increases or exclusions from certain covers. If you refuse the offer, you can renew the policy at the original conditions.

You will now be presented with 5 short scenarios in which your insurance provider requests access to your genomic data. Please read and evaluate carefully each scenario by using the scale provided.

S.1 You are asked to grant consent to allow your de-identified data to be used once you have been made aware of its intended use. Further details about the actual use of your data won't be provided once permission is granted. You will receive your personal results including your risk of contracting specific diseases. No further support will be provided to deal with the results. You will be required to pay the cost of the testing (€250). *Would you share your genomic data?*

S.2 You are asked to grant consent to allow your de-identified data to be used once you have been made aware of its intended use. Further details about the actual use of your data won't be provided once permission is granted. You will receive your personal results including your risk of contracting specific diseases and you will have a free facilitated access to genomic counselling in order to analyze the results and promote disease prevention. You will be required to pay the cost of the testing (€250). *Would you share your genomic data?*

S.3 You are asked to grant consent to allow your de-identified data to be used once you have been made aware of its intended use. Further details about the actual use of your data won't be provided once permission is granted. You will receive your personal results including your risk of contracting specific diseases and you will have a free facilitated access to genomic counselling in order to analyze the results and promote disease prevention. The cost of your genomic test will be covered by the company (€250). *Would you share your genomic data?*

S.4 You are asked to grant consent to allow your de-identified data to be used once you have been made aware of its intended use. Thereafter, you will receive regular updates containing information with regard to which research institutions are analyzing your data, the aim of each project, and the output from previous projects. You will receive your personal results including your risk of contracting specific diseases. No further support will be provided to deal with the results.
You will be required to pay the cost of the testing (€250). *Would you share your genomic data?*

S.5 You are asked to grant consent to allow your de-identified data to be used once you have been made aware of its intended use. Thereafter, you will receive regular updates containing information with regard to which research institutions are analyzing your data, the aim of each project, and the output from previous projects. You will receive your personal results including your risk of contracting specific diseases and you will have a free facilitated access to genomic counselling in order to analyze the results and promote disease prevention. You will be required to pay the cost of the testing (€250). *Would you share your genomic data?*

S.6 You are asked to grant consent to allow your de-identified data to be used once you have been made aware of its intended use. Thereafter, you will receive regular updates containing information with regard to which research institutions are analyzing your data, the aim of each project, and the output from previous projects. You will receive your personal results including your risk of contracting specific diseases and you will have a free facilitated access to genomic counselling in order to analyze the results and promote disease prevention. The cost of your genomic test will be covered by the company (€250).
Would you share your genomic data?

S.7 You are made aware of how your de-identified genomic data is intended to be used. You will be able to grant or deny consent to the use of your data in future research projects depending on your personal preferences. Thereafter, you will receive regular updates containing information with regard to which research institutions are analyzing your data, the aim of each project, and the output from previous projects. You will receive your personal results including your risk of contracting specific diseases. No further support will be provided to deal with the results.

You will be required to pay the cost of the testing (€250).
Would you share your genomic data?

S.8 You are made aware of how your de-identified genomic data is intended to be used. You will be able to grant or deny consent to the use of your data in future research projects depending on your personal preferences. Thereafter, you will receive regular updates containing information with regard to which research institutions are analyzing your data, the aim of each project, and the output from previous projects. You will receive your personal results including your risks of contracting specific diseases and you will have a free facilitated access to genomic counselling in order to analyze the results and promote disease prevention. You will be required to pay the cost of the testing (€250).
Would you share your genomic data?

S.9 You are made aware of how your de-identified genomic data is intended to be used. You will be able to grant or deny consent to the use of your data in future research projects depending on your personal preferences. Thereafter, you will receive regular updates containing information with regard to which research institutions are analyzing your data, the

aim of each project, and the output from previous projects. You will receive your personal results including your risk of contracting specific diseases and you will have a free facilitated access to genomic counselling in order to analyze the results and promote disease prevention. The cost of your genomic test will be covered by the company (€250).
Would you share your genomic data?

S.10 You are made aware of how your de-identified genomic data is intended to be used. You will be able to grant or deny consent to the use of your data in future research projects depending on your personal preferences. You will also be able to give an automatic consent for particular research topics and receive updates on the progress of projects. You will receive your personal results including your disease risks from the company's network. No further support will be provided to deal with the results.

You will be required to pay the cost of the testing (€250).
Would you share your genomic data?

S.11 You are made aware of how your de-identified genomic data is intended to be used. You will be able to grant or deny consent to the use of your data in future research projects depending on your personal preferences. You will also be able to give an automatic consent for particular research topics and receive updates on the progress of projects. You will receive your personal results including your risk of contracting specific diseases and you will have a free facilitated access to genomic counselling in order to analyze them and promote disease prevention.
You will be required to pay the cost of the testing (€250).
Would you share your genomic data?

S.12 You are made aware of how your de-identified genomic data is intended to be used. You will be able to grant or deny consent to the use of your data in future research projects depending on your personal preferences. You will also be able to give an automatic consent for particular research topics and receive updates on the progress of projects. You will receive your personal results including your risk of contracting specific diseases and you will have a

free facilitated access to genomic counselling in order to analyze the results and promote disease prevention. The cost of your genomic test will be covered by the company (€250).

Would you share your genomic data?